THE DUAL OF A PERFECT COMPLEX IS PERFECT AND REFINES THE DOUBLE DUAL AND THE TENSOR-HOM ADJUNCTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes '(this uses Lemma ... to see that E^bullet is the left dual of K^bullet in the category of complexes)' inside the counted proof span, reproduced verbatim. The nine references inside the counted proof span are recorded as non-load-bearing navigation pointers; they include cross-chapter tags and same-chapter label lines.

Allowed standard prerequisites (editorial).

- Bounded complexes of finite projective modules representing perfect objects. used at proof lines 19189-19190. Is the representation on which the explicit computation is carried out.
- The internal Hom complex and the evaluation map up to sign. invoked at proof lines 19196-19202. Supplies the canonical map whose bijectivity gives the double dual isomorphism.
- Left duals in the category of complexes. invoked at proof lines 19220-19222. Is what turns the sign-compatible isomorphism into the tensor-Hom adjunction.

Source reference. Original lemma 064B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7860–7860. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 06XP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8858–8858. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 18533: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18533–18533. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 18708: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18708–18708. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18164: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18164–18164. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18682: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18682–18682. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18462: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18462–18462. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original section 18339: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18339–18339. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a ring. Let $K \in D(A)$ be perfect. Then $K^\vee = R\mathrm{Hom}_A(K, A)$ is a perfect complex and $K \cong (K^\vee)^\vee$. There are functorial isomorphisms*

$$L \otimes^{\mathbf{L}}_A K^\vee = R\mathrm{Hom}_A(K, L) \quad \text{and} \quad H^0(L \otimes^{\mathbf{L}}_A K^\vee) = \mathrm{Ext}^0_A(K, L)$$

for $L \in D(A)$.

Proof. We can represent K by a bounded complex $K^\bullet$ of finite projective A-modules. By Lemma 18708 the object $K^\vee$ is represented by the complex $E^\bullet = \mathrm{Hom}^\bullet(K^\bullet, A)$. Note that $E^n = \mathrm{Hom}_A(K^{-n}, A)$ is finite projective as well. Thus $E^\bullet$ is a bounded complex of finite projective modules and we conclude $K^\vee$ is perfect. There is a canonical map

$$K^\bullet = \mathrm{Tot}(\mathrm{Hom}^\bullet(A, A) \otimes_A K^\bullet) \longrightarrow \mathrm{Hom}^\bullet(\mathrm{Hom}^\bullet(K^\bullet, A), A)$$

which up to sign uses the evaluation map in each degree, see Lemma 18164. (For sign rules see Section 18339.) Thus this map defines a canonical isomorphism $(K^\vee)^\vee \cong K$ as the double dual of a finite projective module is itself.

The second equality follows from the first by Lemma 18682 and Derived Categories, Lemma 064B as well as the definition of Ext groups, see Derived Categories, Section 06XP. Let $L^\bullet$ be a complex of A-modules representing L. By Section 18339 item (18533) there is a canonical isomorphism

$$\mathrm{Tot}(L^\bullet \otimes_A E^\bullet) \longrightarrow \mathrm{Hom}^\bullet(K^\bullet, L^\bullet)$$

of complexes of A-modules (this uses Lemma 18462 to see that $E^\bullet$ is the left dual of $K^\bullet$ in the category of complexes). This proves the first displayed equality and the proof is complete. □